



Autocorrelation and Moran'I

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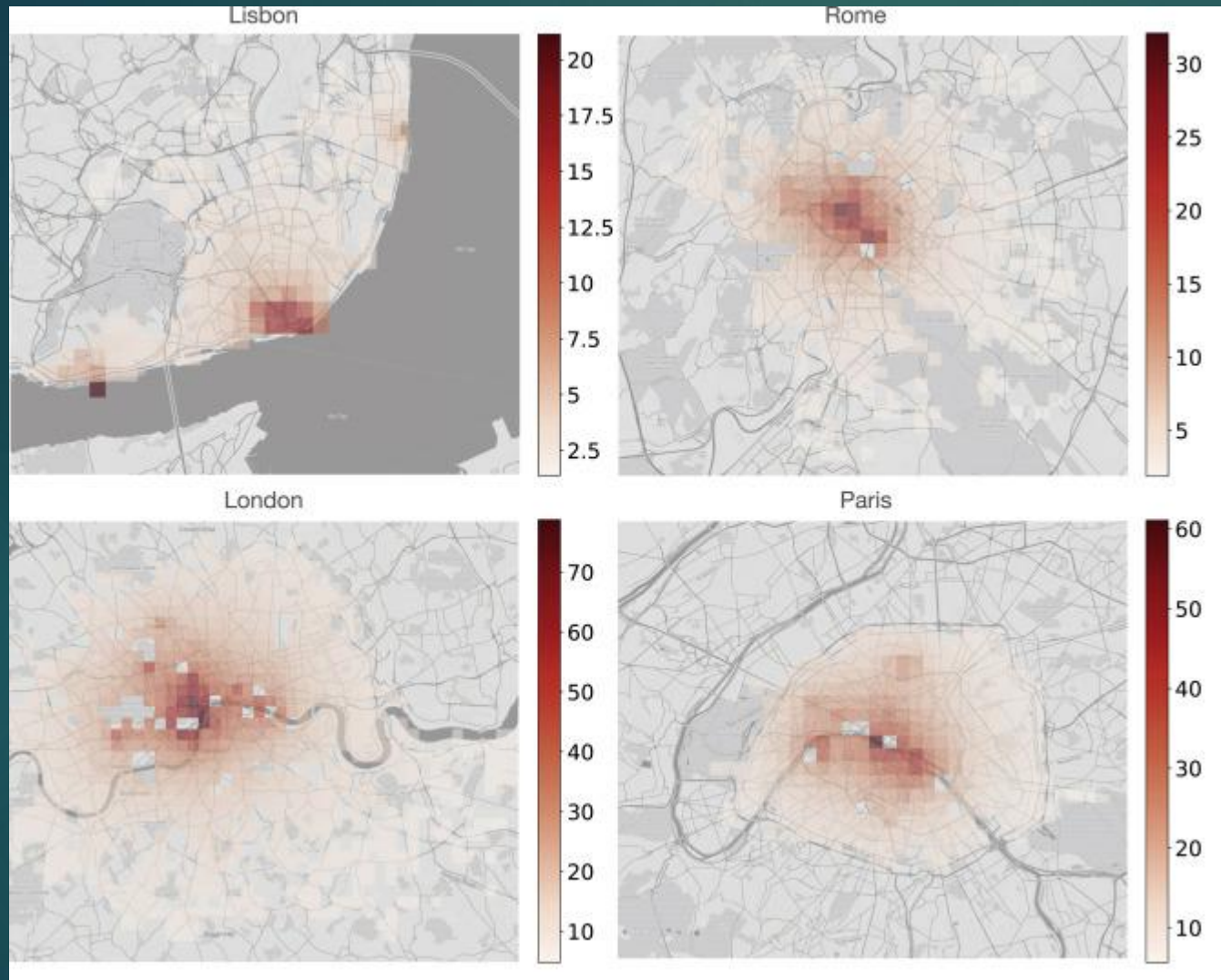
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Content

1. What is Spatial Autocorrelation?
2. What is Moran'I?
3. How to compute Moran'I?
4. Applications of Autocorrelation
5. Practice

Comment on the plot

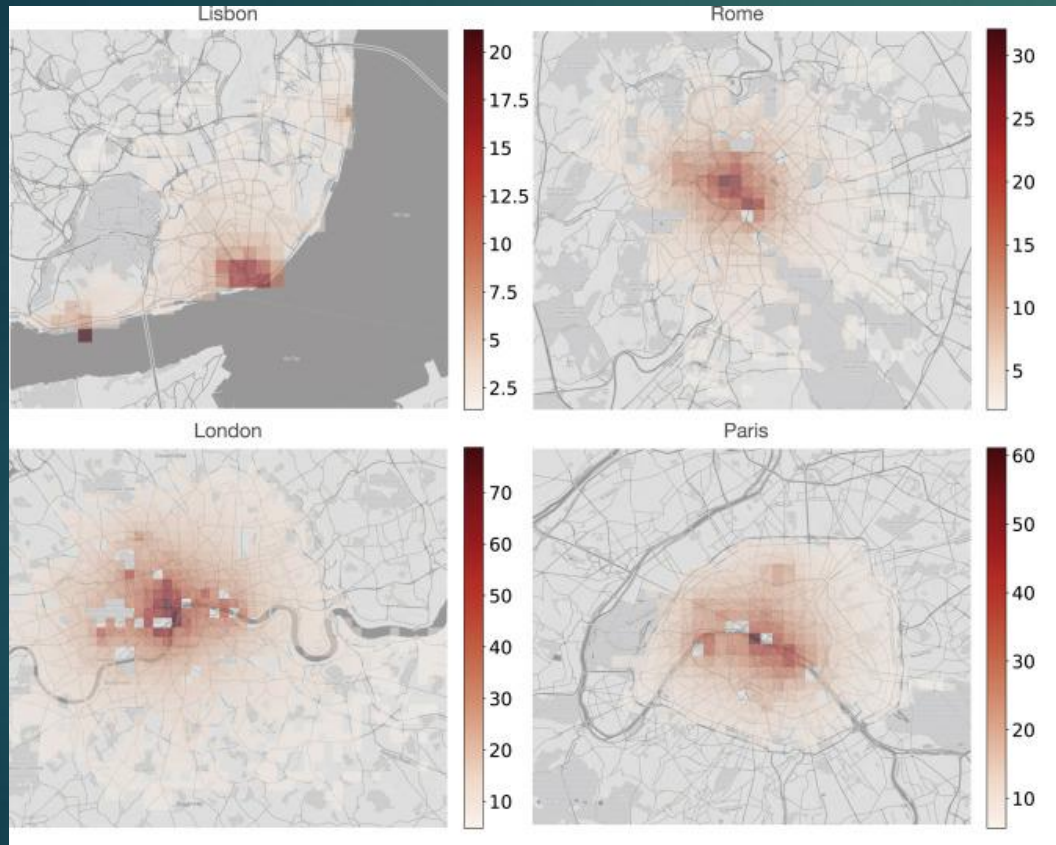


Attraction index in selected cities

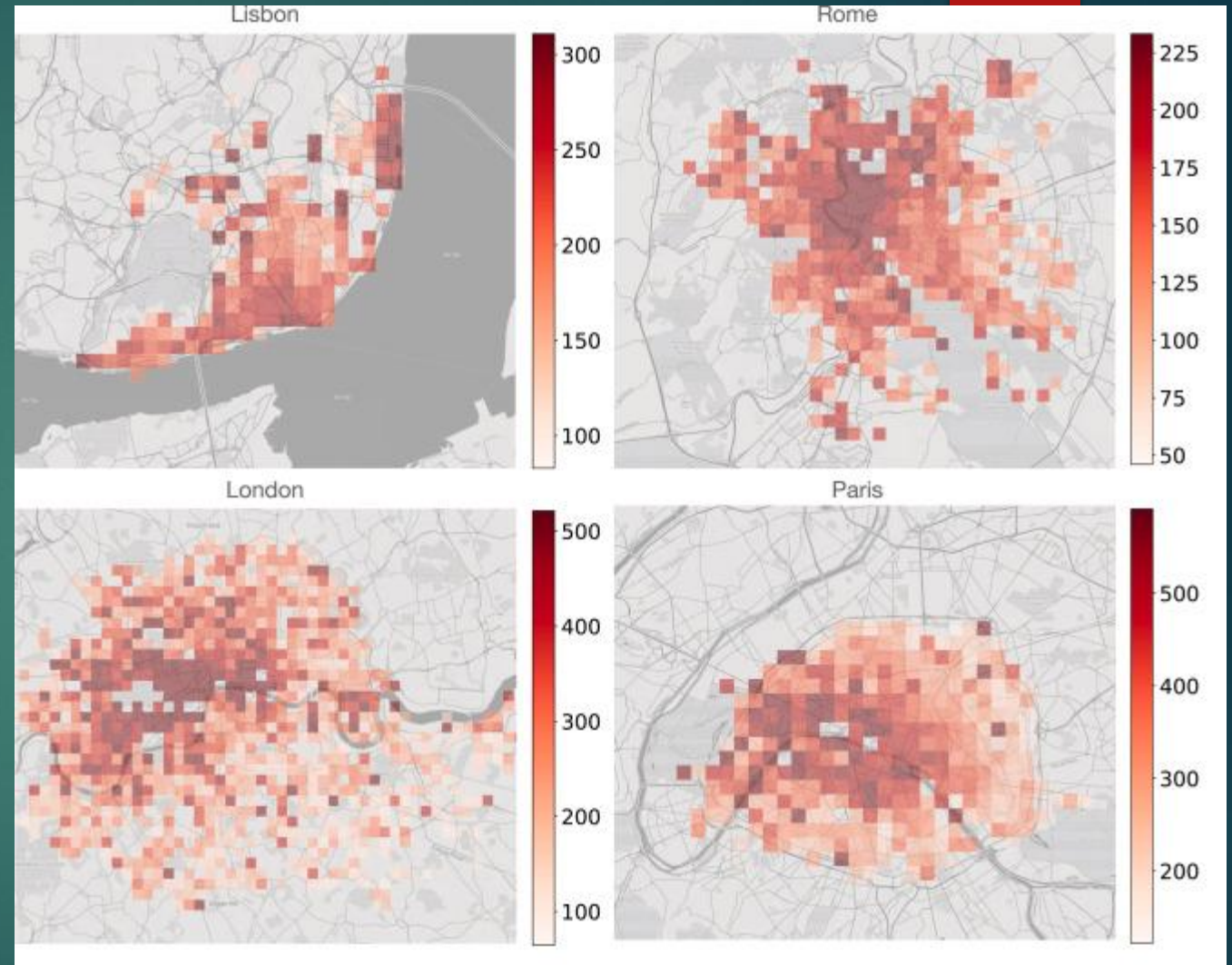
The Fig. presents heatmaps for mean values of `attr_index`.

The maps show the concentration of attractions in the city centre: in general, the listings in central areas have a higher accessibility of attractions; while further from the centre, the index values decline.

Comment on the linked plots



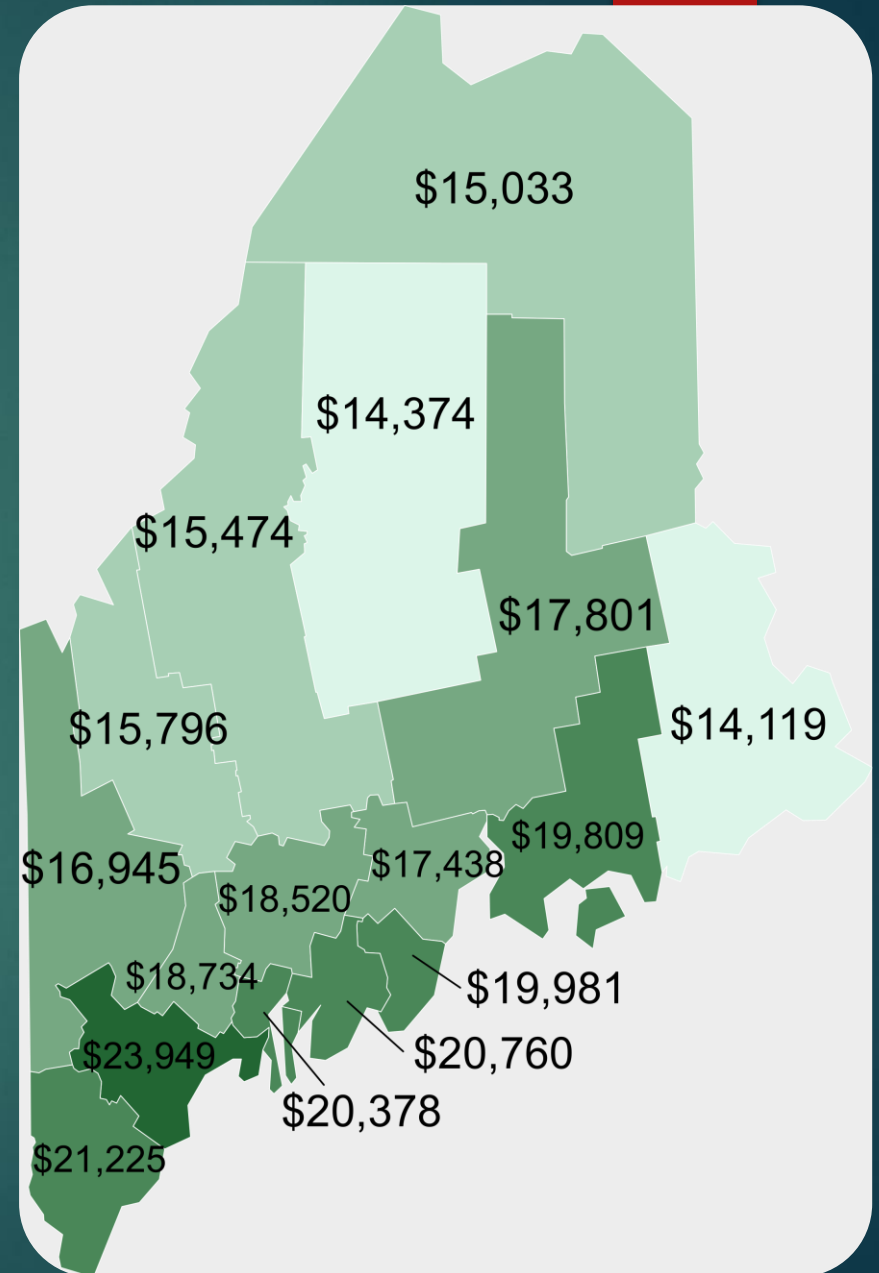
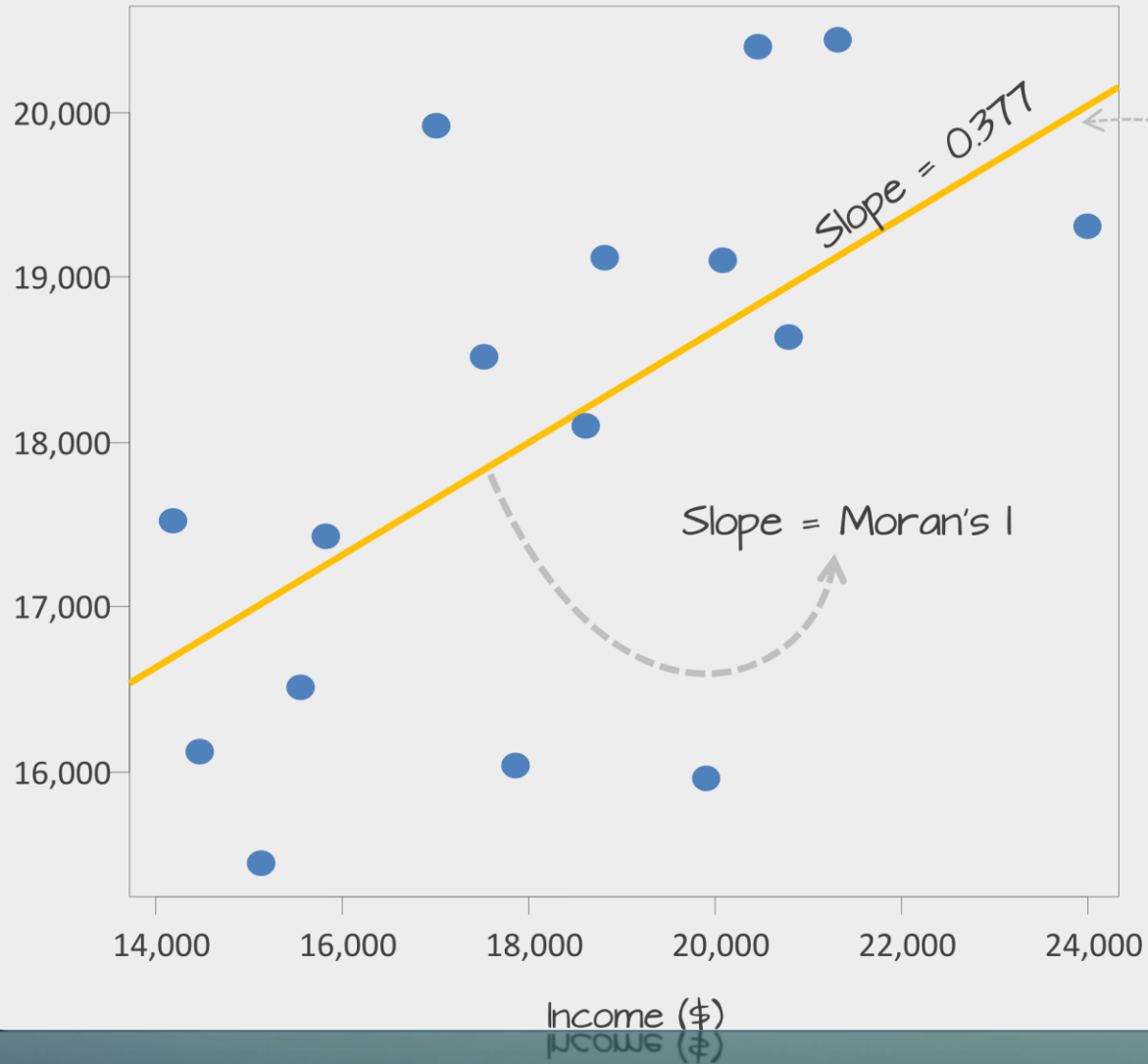
Attraction index in selected cities



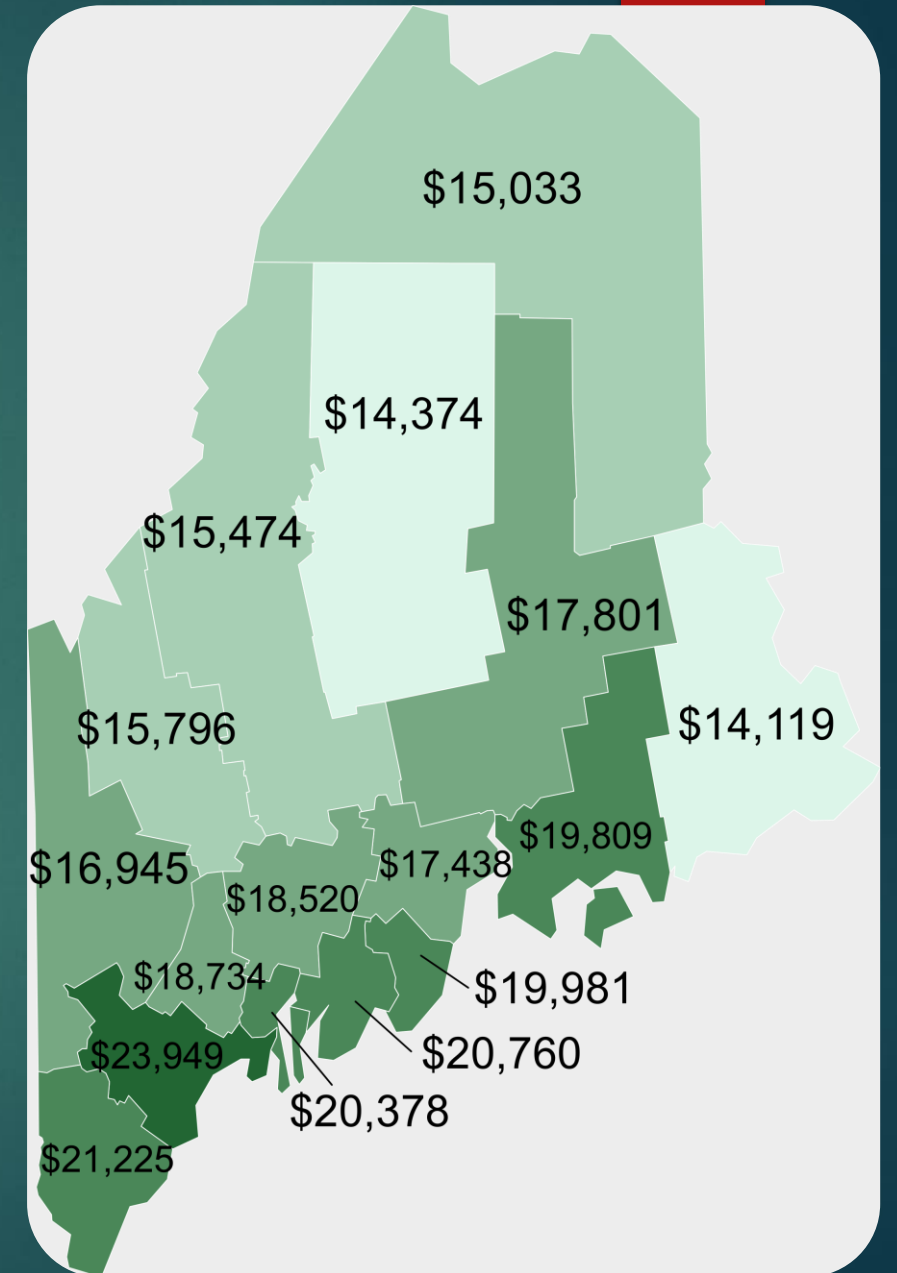
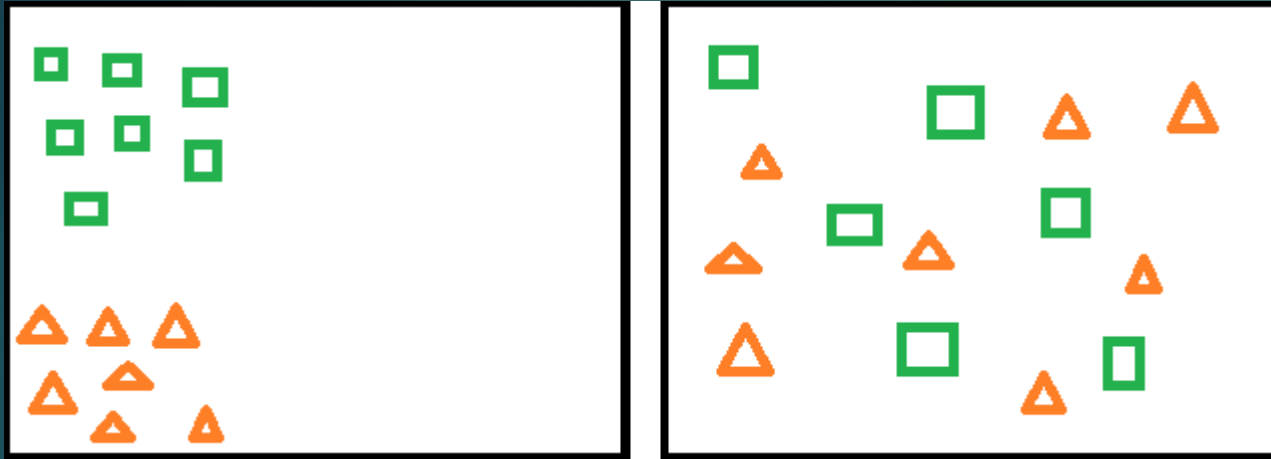
Mean Airbnb prices (in EUR).

How to quantify autocorrelation? (The Moran's I)

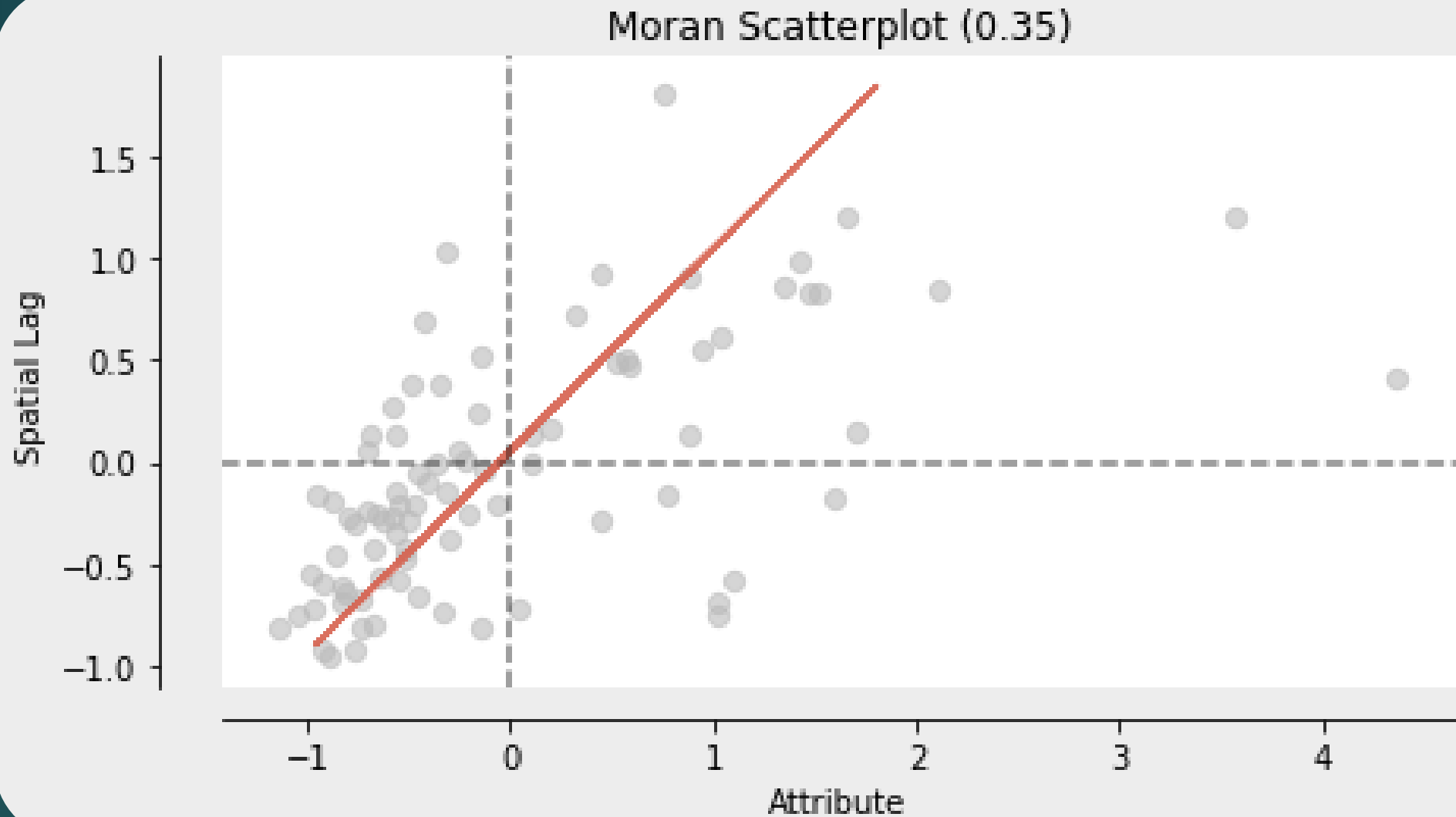
Neighboring (mean) income (\$)



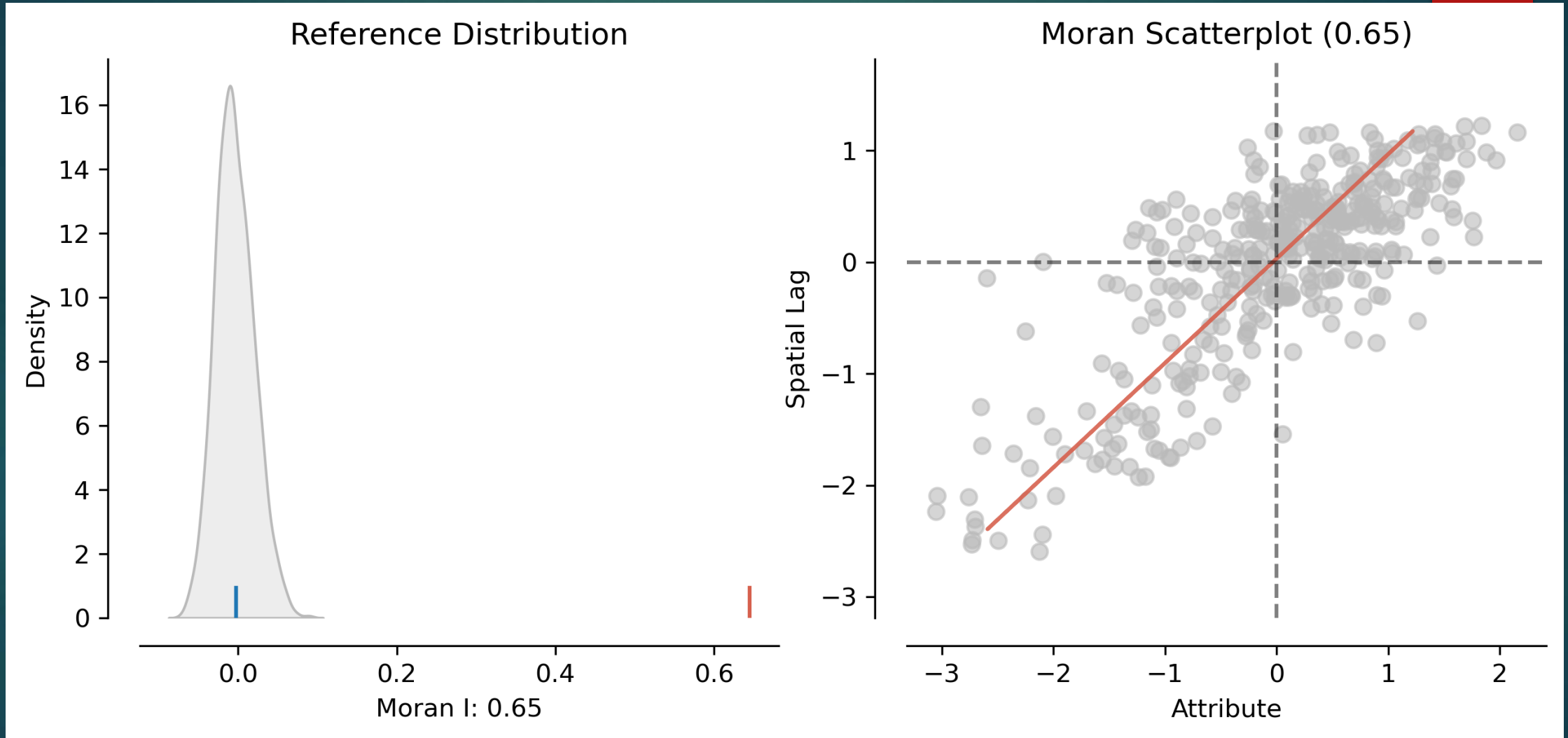
Comment on the below plots



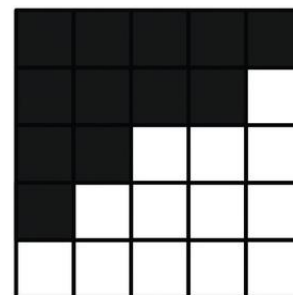
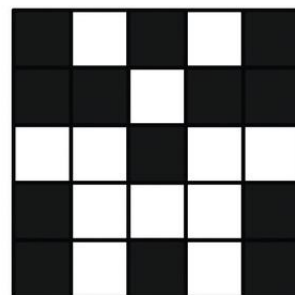
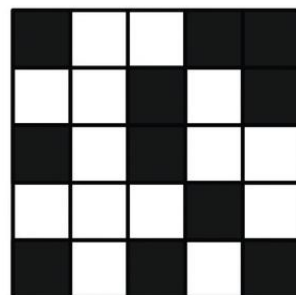
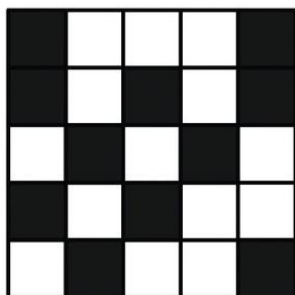
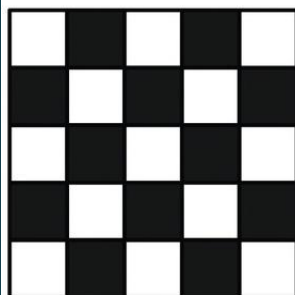
Understanding the Moran's I plot



Practice



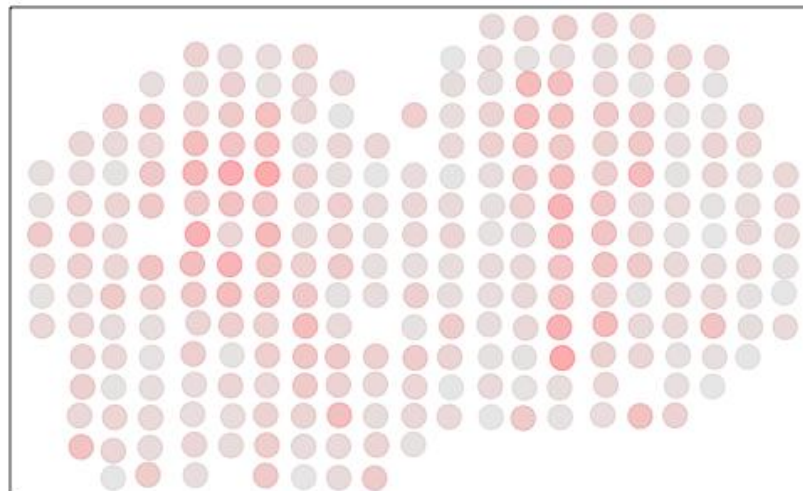
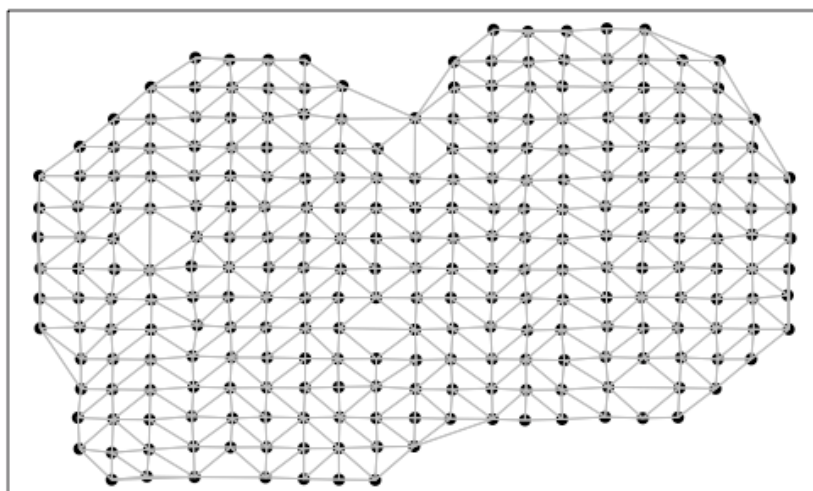
Moran's I



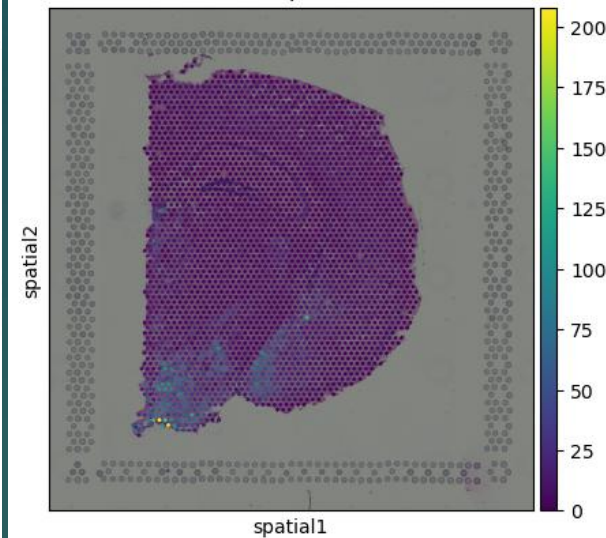
Moran's $I < E(I)$
indicates tend
to dispersion

Random
Moran's $I = E(I)$

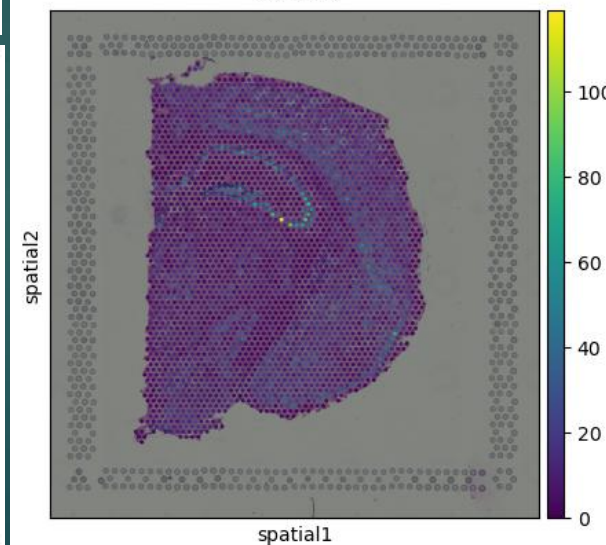
Moran's $I > E(I)$
indicates tend
to clustering



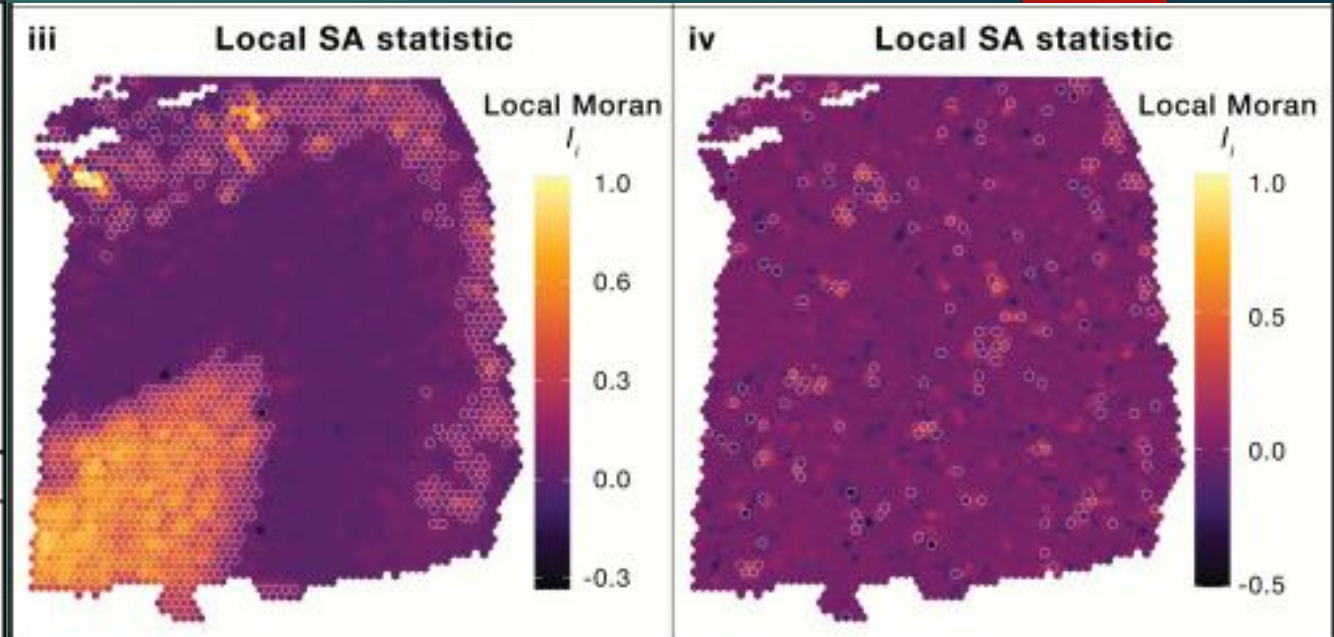
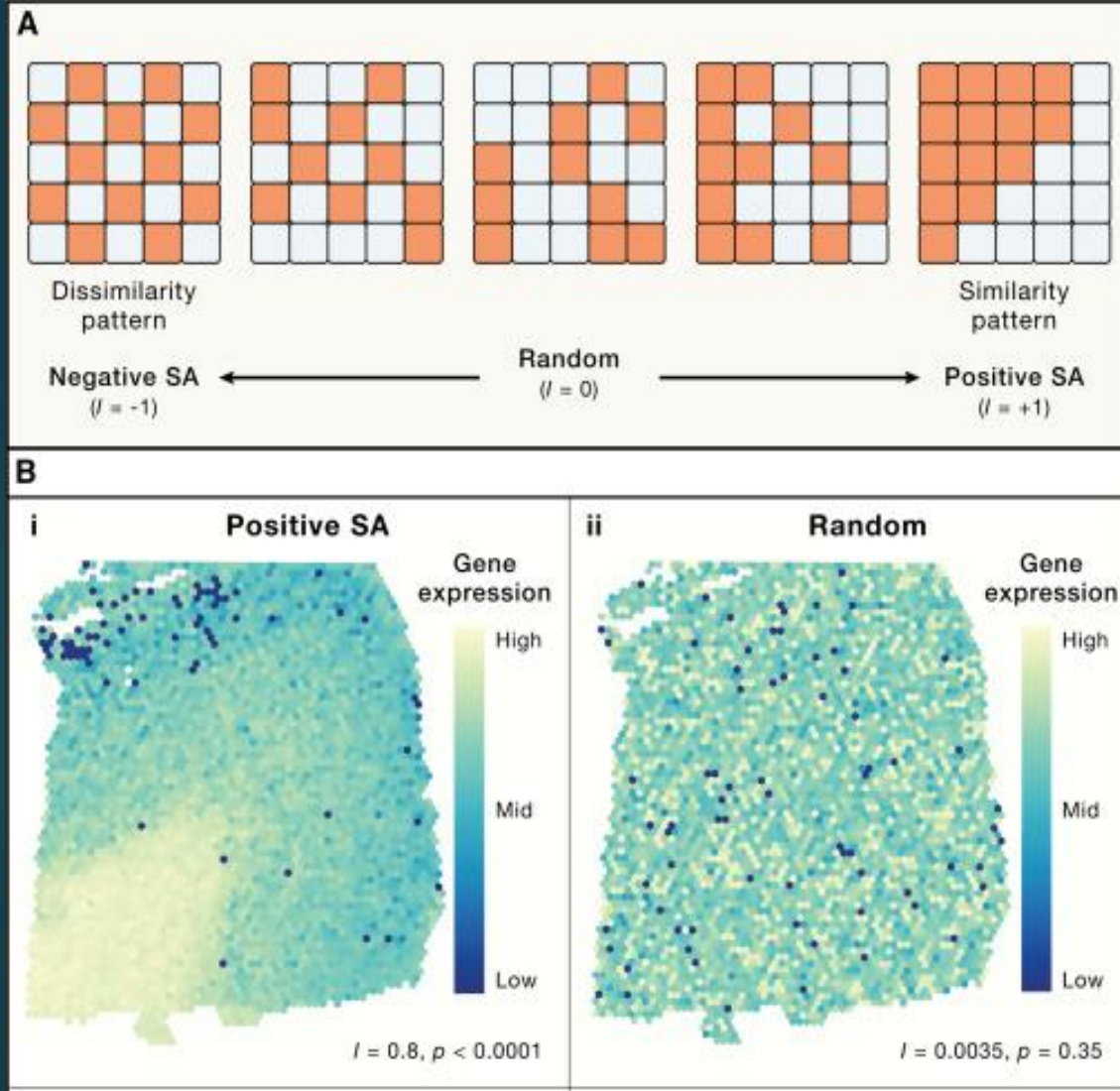
Resp18



Tuba4a



Spatial autocorrelation highlights regions of statistical similarity in gene-wise expression



(A) A simulated example of spatial autocorrelation (SA) for binary data associated with Moran's I statistics. The leftmost panel shows perfect dispersion or dissimilarity, corresponding to a Moran's I of -1 . The center panel shows a random spatial distribution, corresponding to Moran's I of 0 . The rightmost panel shows perfect spatial autocorrelation, clustering, or similarity, corresponding to a Moran's I of $+1$.

(B) Examples of SA from tissue measurements of gene expression. (i) Positive SA ($I = 0.8$): notice how high values cluster in one corner of the tissue while low values are highly localized. (ii) The same observations but randomly distributed over space ($I \approx 0$). (iii) The local Moran's I_i (LISA) for the variable in (i). Here, the local statistic provides a more detailed representation of SA over space, revealing locations with high SA. Significant locations ($p < 0.05$) are highlighted with a white border. (iv) The local significance of the local Moran's I_i (LISA) in (ii). Significant locations ($p < 0.05$) are highlighted with a white border.

Example applications

The following are example applications of the tool:

- Identify an appropriate neighborhood distance for a variety of spatial analysis methods by finding the distance where spatial autocorrelation is strongest.
- Measure broad trends in ethnic or racial segregation over time—is segregation increasing or decreasing?
- Summarize the diffusion of an idea, disease, or trend over space and time—is the idea, disease, or trend remaining isolated and concentrated, or spreading and becoming more diffuse?



Xin chân thành cảm ơn!

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